

SOUTHWEST MICHIGAN REGIONAL, MI, USA

WMO: 726355

Lat: 42.126N

Lon: 86.428W

Elev: 196

StdP: 98.99

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 94871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-12.5	-18.9	0.7	-14.3	-16.8	0.9	-12.0	12.1	-6.2	11.2	-5.2	4.0	180	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	32.0	23.6	30.1	22.6	28.6	22.0	24.9	29.8	24.0	28.4	23.1	27.1	4.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.2	27.4	22.5	17.6	26.8	22.0	17.1	26.1	76.5	29.9	72.9	28.4	69.5	27.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	9.0	8.1	-19.6	34.5	3.9	1.8	-22.4	35.8	-24.7	36.8	-26.9	37.8	-29.8	39.2		
			-20.0	26.6	3.7	1.2	-22.7	27.5	-24.8	28.2	-26.9	28.8	-29.6	29.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-3.1	-1.9	2.5	8.5	14.2	19.3	21.7	20.8	17.4	11.3	5.2	-0.1	(6)
	DBStd	9.98	5.84	5.44	5.96	5.26	5.15	4.29	3.57	3.10	4.29	4.85	4.91	5.11	(7)
	HDD10.0	1612	406	334	247	91	17	1	0	0	1	41	158	316	(8)
	HDD18.3	3489	664	566	493	299	151	39	10	12	68	223	394	571	(9)
	CDD10.0	1509	1	1	13	45	146	280	363	335	224	83	15	3	(10)
	CDD18.3	345	0	0	1	3	22	68	115	88	41	7	0	0	(11)
	CDH23.3	2963	0	0	4	31	216	621	1041	677	324	49	0	0	(12)
	CDH26.7	849	0	0	0	4	56	191	332	171	87	7	0	0	(13)
(14) Wind	WSAvg	3.5	4.5	4.2	3.9	3.9	3.3	2.8	2.5	2.4	2.7	3.5	4.1	4.4	(14)
(15) Precipitation	PrecAvg	894	59	46	50	75	86	89	84	102	89	92	80	58	(15)
	PrecMax	1149	125	110	106	132	169	194	182	277	280	221	184	144	(16)
	PrecMin	696	15	13	13	8	18	26	20	37	22	12	28	25	(17)
	PrecStd	130	29	23	26	34	44	47	38	56	59	53	38	28	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.4	23.9	27.1	31.0	32.9	34.1	32.8	31.9	27.6	20.9	16.3	(19)
		MCWB	12.9	10.5	16.9	17.2	21.5	23.4	25.0	24.4	23.0	20.8	14.3	13.9	(20)
	2%	DB	10.3	12.2	19.2	23.4	28.1	31.1	32.1	30.5	29.0	24.2	17.5	12.6	(21)
		MCWB	8.2	9.0	14.2	15.5	19.8	22.5	24.3	23.0	22.0	17.8	13.1	10.3	(22)
	5%	DB	7.2	7.9	15.9	21.0	25.9	28.8	30.0	28.4	27.2	22.1	15.3	9.1	(23)
		MCWB	5.1	5.3	11.9	13.7	18.7	21.6	23.1	22.3	20.7	17.4	11.6	7.1	(24)
	10%	DB	4.4	5.7	12.2	17.8	23.1	27.2	28.1	27.0	24.8	19.3	12.8	6.8	(25)
		MCWB	2.9	3.3	8.7	12.3	17.1	20.8	22.4	21.7	19.7	15.6	9.6	4.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.3	17.5	19.2	23.0	25.1	26.5	25.7	24.2	22.2	15.9	14.7	(27)
		MCDB	14.3	15.5	22.7	25.0	29.8	31.2	32.0	30.7	29.1	26.5	18.6	16.2	(28)
	2%	WB	8.6	9.0	15.1	16.7	21.0	23.8	25.3	24.4	22.9	19.8	14.2	10.9	(29)
		MCDB	9.9	11.8	18.8	21.2	26.3	28.8	30.0	28.7	27.1	22.8	17.0	12.2	(30)
	5%	WB	5.0	5.8	12.0	14.8	19.6	22.7	24.3	23.5	21.8	17.9	12.1	7.5	(31)
		MCDB	6.7	8.2	15.0	19.5	24.1	27.3	28.7	27.1	25.4	20.8	14.7	8.6	(32)
	10%	WB	3.0	3.2	8.6	12.7	18.1	21.6	23.3	22.5	20.6	16.0	10.1	4.9	(33)
		MCDB	4.4	5.2	11.8	17.7	22.2	25.8	27.3	25.7	24.0	19.3	12.7	6.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.2	8.9	10.9	11.1	10.8	10.4	10.2	10.9	9.7	7.5	6.1	(35)
		MCDBR	9.2	10.9	13.5	15.1	13.7	12.3	11.9	12.0	12.8	12.9	11.0	9.2	(36)
	5% WB	MCWBR	7.8	8.8	9.5	8.9	7.4	6.0	5.6	6.1	6.6	7.9	8.0	8.0	(37)
		MCWBR	8.7	10.4	12.3	13.3	12.0	10.6	10.4	9.7	10.3	10.8	9.6	8.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.303	0.326	0.345	0.390	0.419	0.430	0.437	0.434	0.407	0.357	0.321	0.305	(40)
		taud	2.427	2.351	2.370	2.278	2.258	2.276	2.258	2.267	2.307	2.441	2.489	2.468	(41)
	Ebn at Noon		843	873	899	878	858	847	836	827	822	829	816	809	(42)
		Edh at Noon	77	98	108	128	134	132	133	128	115	89	73	69	(43)
(44) All-Sky Solar Radiation	RadAvg		1.32	2.12	3.55	4.74	5.44	6.20	6.29	5.41	4.36	2.77	1.65	1.10	(44)
	RadStd		0.14	0.29	0.29	0.40	0.49	0.40	0.29	0.31	0.37	0.31	0.25	0.14	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page